

EarWiggl'In: The Ear Wiggling Input Method for Smart Glasses using Pressure Sensors on Nose Pads

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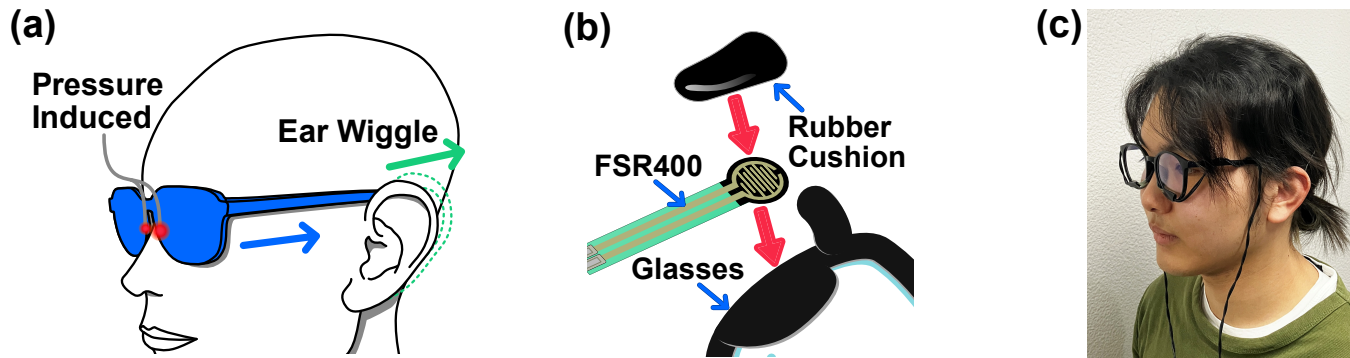


Figure 1: EarWiggl'In. (a) Glasses moving corresponded with ear wiggling, inducing pressure on the nose pads. (b) Pressure sensor(FSR400) and the rubber cushion attached to the nose pad of the glasses. (c) A user wearing the device.

Abstract

Eye trackers on smart glasses enable cursor control, but an additional input trigger is required for target selection. EarWiggl'In introduces a hands-free and eyes-free input system for smart glasses that utilize pressure sensors on the nose pads. The system detects input by measuring the pressure induced when the glasses are pulled back with the ear wiggling. A user study was conducted to evaluate the input speed of ear wiggling, eight ear wiggle gestures, and the workload using NASA-RTLX. Results indicated that ear wiggling input speed was comparable to, though slightly slower than, pressing a button with a finger, with an average delay of 36.28[ms]. Additionally, eight gestures were analyzed, revealing different amplitudes, numbers of peaks, and durations depending on the gesture.

CCS Concepts

• **Human-centered computing** → **Interaction devices; Mobile computing; Gestural input; Human computer interaction (HCI).**

Keywords

Wearable Devices, Smart Glasses, Hands-Free Input, Eyes-Free Input, Ear Movement, Ear Wiggle Input

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1 Introduction

Eye trackers on smart glasses provide gaze input and enable cursor control. However, an additional input trigger is necessary for target selection to fully operate the device. Dwell-based methods [18, 30] are commonly used, but they are prone to the Midas Touch problem [15]. Alternative solutions use eye gestures [4, 21, 26] or other human movements [7, 10, 24] as input triggers to prevent the problem. Nevertheless, these inputs take a certain duration and may interfere with daily movements.

When designing a practical input for smart glasses, it is essential to avoid requiring hand-held or other worn devices, prevent increasing the device weight, minimize the interference with daily activities, and ensure usability in various environments (e.g. light conditions, acoustic conditions). To address these challenges, this research focuses on using ear wiggling, a movement that does not interfere with daily movements. Ear wiggling provides a fast and voluntary input as a mouse click. Based on this concept, I developed EarWiggl'In, an input method for smart glasses by utilizing pressure sensors on the nose pads. This approach is resistant to environmental factors and is designed to be lightweight.

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In this paper, I examined the input speed and workload associated with using ear wiggling as an input trigger. First, I conducted a user study to compare the proposed method with finger-based button pressing. Following the measuring, the participants completed the NASA Raw Task Load Index(NASA-RTLX) questionnaire. Additionally, I analyzed various ear-wiggling gestures, as prior research on ear-wiggling gestures was limited. Participants were instructed to perform the eight gestures and subsequently completed the NASA-RTLX questionnaire.

2 Related Works

EarWiggl'In is a hands-free input method for smart glasses that utilizes pressure sensors to detect ear wiggling. This section provides an overview of prior research on hands-free input methods and ear wiggling detection.

2.1 Hands-Free Input Methods for Eye-Trackers

Dwell-based methods [18, 30] are commonly used input trigger, as they do not require additional equipment. Their effectiveness can vary depending on the dwell time, with expert users achieving up to 300[ms] of response time by adjustable this parameter[18]. Gaze gestures [4, 21, 26] are also a common method for target selection. Pursuits [29] enables users to select moving targets by tracing the movement with their eyes. Comparison between these methods and dwell-based approaches resulted preferences that vary in different situations[22]. However, short duration or simple gestures can lead to the Midas Touch problem [15], while long duration or complex gestures may reduce efficiency. Additionally, gaze gestures often draw too much user attention, and Pursuits is ineffective for non-moving targets.

Utilizing other movements such as hand movement [7], head movement [24], tongue gesture [10], has also been studied as an input trigger. Compared to dwell-based methods, these methods are faster and effectively mitigate Midas Touch. However, these methods overlap with actions performed in daily life, causing interfering with daily movement and causing misoperation. In contrast, EarWiggl'In leverages ear wiggling as a hands-free input trigger. Since ear wiggling is rarely used in daily life, it provides a clear and intentional input method that does not disrupt daily activities.

2.2 Ear Wiggling and Detection

Ear wiggling refers to the movement of the ear driven by the auricular muscle, which are generally considered to be vestigial [9]. Despite this, 95% of the population still possess the muscles [11] and nerves [3, 23], with 10–20% of individuals capable of voluntary ear wiggling [8, 14, 17]. Research has shown that individuals can regain control of these muscles through training [6, 16, 27].

Electromyograms(EMG) are used in various fields to detect the auricular muscle movements [6, 16, 25, 27]. As for use cases, Reach Bionics [1, 28] developed a headset for controlling a device, and Aoyama et al. [27] used ear wiggling to control a robotic thumb. Similarly, InEar BioFeedController [19] employed electroencephalography sensors embedded in in-ear headphones to detect ear wiggling as an input. While EMGs have a high sensitivity and can detect minute movements, require skin preparation and precise electrode placement, and are prone to detachment during extended



Figure 2: Pressure sensors being attached to the nose pad for the EarWiggl'In device

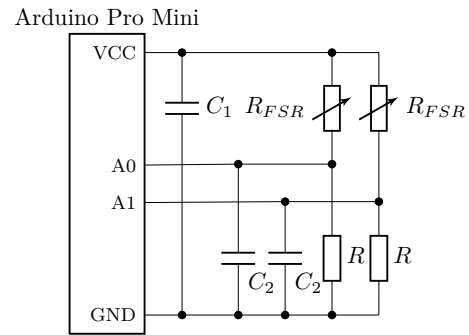


Figure 3: Circuit of the EarWiggl'In device

use, making them unsuitable for wearable devices. Bair [2] utilized kymographs for similar purposes, but the device's size and lack of digital measurement is not practical for wearable use.

In contrast, EarWiggl'In detects ear wiggling using pressure sensors. This method eliminates the need for skin preparation or precise setup before use and provides a stable contact for continuous operation. Furthermore, the system is compact, lightweight, and easily integrated to a device, making it a practical solution for wearable applications.

3 Sensing Principle of EarWiggl'In

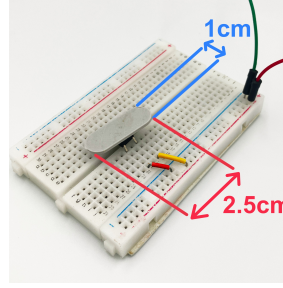
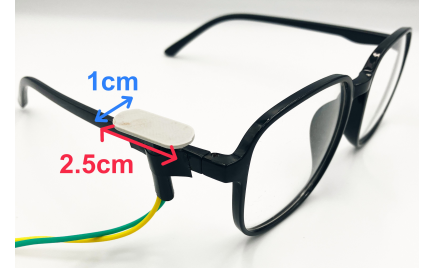
The sensing principle of EarWiggl'In is illustrated in Fig. 1(a). When a user wears glasses and wiggles their ear, the glasses respond to the movement as the outer ear moves back and forth. This motion causes the glasses to be pulled backward, increasing the pressure between the nose pad and the nose. These pressure fluctuations make the pressure differ from the static pressure experienced when simply wearing the glasses. By embedding sensors in the nose pads, this mechanism translates ear wiggling into an input trigger.

4 Implementation

The EarWiggl'In method was implemented on a pair of glasses. Glasses with nose pads integrated into the frame were chosen to avoid interference from deformable components by the nose wires. Pressure sensors (Interlink Electronic, FSR400) were attached to

Table 1: Participant ear wiggling skills

	Both	Left Only	Right Only
P1	✓		
P2	✓	✓	✓
P3	✓		
P4	✓		✓
P5	✓	✓	✓

**Figure 4: Mouse Click Device****Figure 5: On-Button Device**

each nose pad, with silicone rubber cushions added on top to enhance sensitivity and stability (Fig. 1(b), Fig. 2). The sensors were connected to a circuit mounted on a breadboard, powered and measured by an Arduino Pro Mini (Fig. 3). The Arduino measures the voltage of resistor R , which reflects changes in resistance when pressure is applied to the sensors. Capacitor C_1 is a bypass capacitor, while capacitor C_2 was included to reduce noise signals. The component values were set as follows: $R = 7.5[\text{k}\Omega]$, $C_1 = C_2 = 0.1[\mu\text{F}]$. The Arduino transmitted the recorded values, including timestamps and pressure values, to a laptop computer, where they were saved in a CSV file. The system sampling rate was 100[Hz]. The maximum power consumption of the measuring part is 6.7[mW] and the weight of the device was 18.8[g].

5 Evaluation

A user study was conducted to evaluate the input speed and the workload associated with using ear wiggling as an input trigger. Five participants (P1–P5; average age = 19.40 years, SD = 2.64; all male and right-handed), were recruited from the local institution. All participants were capable of voluntarily wiggling their ears, as visually confirmed by the author. The participant’s ability to move their ears is summarized in Table 1. P1 stated that he could wiggle each of the ears independently in the past. P2 mentioned experiencing involuntary ear wiggling in situations requiring intense focus, while other participants did not state such occurrences. P2, P3, P4 and P5 stated that they occasionally wiggle their ears in daily life, such as when relaxing his head, adjusting slipped glasses, pulling glasses closer to look at distant objects, or adjusting his glasses when his hands are occupied. All participants stated that they never experienced pain during or after ear wiggling in the past.

The experiment was conducted under sitting conditions. First, the procedure was explained to the participants, followed by a questionnaire to gather demographic information and their experiences with ear wiggling. Two experiment tasks, Task A and Task B were then performed, as described in later sections. During both tasks, the participants completed the NASA Raw Task Load Index(NASA-RTLX) [5, 13] questionnaire, with Japanese titles and descriptions in the study by Haga and Mizukami[12]. A five-minute break was given between the two tasks. After completing both tasks, they filled out an additional questionnaire regarding tiredness and comments on the input method. The entire experiment

took approximately 90 minutes per participant. All participants were paid JPY2000 (approximately USD12.8) with QUO cards.

Task A. The purpose of Task A was to compare the input speed and workload between EarWiggl’In and conventional input. Participants performed three types of input methods: Mouse Click, On-Button, and EarWiggl’In. To equalize system latency across methods, tact switches were implemented as in Fig. 4 and Fig. 5 for Mouse Click and On-Button. The Mouse Click simulated a typical computer mouse input, representing a common interaction in daily life. On-Button mimicked the capture button of the Ray-Ban Meta glasses [20], representing an input method designed for smart glasses. A 1[cm] × 2.5[cm] plastic plate was glued onto the switch to make it easier to press. These tact switches were connected directly to a resistor, and the inputs were detected by the same Arduino Pro Mini used for EarWiggl’In.

Participants were instructed to perform the input as soon as an 11[cm] radius red circle appeared on a black full-screen display. The circle appeared at random intervals between 2 and 6 seconds. To ensure the measurement focused solely on input speed, the participants were asked to keep their finger in ready position on the button for both the Mouse Click and On-Button. The response time between the circle’s appearance and input was recorded.

Each input method included a practice session and a recording session. Both sessions involved performing the input five times. After completing both sessions, the participants filled out the NASA-RTLX questionnaire. The order of the input methods were randomized for each participant, and a two-minute break was provided between sessions to minimize fatigue.

Task B. The purpose of Task B was to analyze the waveforms caused by different ear wiggling gestures and to compare difficulty levels of these gestures. Participants performed the following gestures: “1 wiggle”, “2 wiggle”, “weak 1 wiggle”, “weak 2 wiggle”, “left wiggle”, “right wiggle”, “hold wiggle”, and “mash wiggling”. The gestures “1 / 2 / weak 1 / weak 2 / hold wiggle” and “mash wiggling” are gestures that move both ears simultaneously, while “left / right wiggle” are done by each ear independently. “Left / right wiggle” were done even for the participants who stated that they could not wiggle their ears independently. The gestures “2 / weak 2 wiggle” were designed to mimic a double-click of a mouse. “weak 1 / weak 2 wiggle” were instructed to do weaker than the maximum capability. For “hold wiggle” participants maintained their ears in a pulled

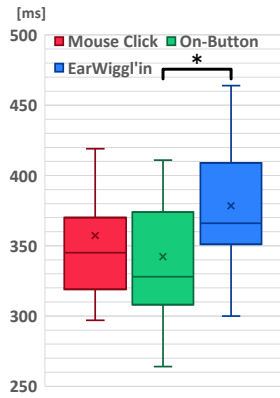


Figure 6: Input speed upon methods(* < 0.05)

back position for 2 seconds. “Mash wiggling” required participants to ear wiggle rapidly as much as possible for 2 seconds.

Each gesture included a practice session and a recording session. During the practice session, participants rehearsed the gesture a few times. In the recording session, the participants performed each gesture 15 times. However, “mash wiggling” was performed only 5 times, due to its higher physical demand than the other gestures. After completing both sessions for each gesture, the participants filled out the NASA-RTLX questionnaire. The order of the gestures was randomized for each participant, and a two-minute break was provided between the gestures to reduce fatigue.

Results

The input time recorded during Task A is shown in Fig. 6. The average input time for Mouse Click was 357.28[ms](SD = 40.02[ms]), for On-Button it was 342.20[ms](SD = 49.86[ms]), for EarWiggl'In it was 378.48[ms](SD = 46.77[ms]). A two-sided t-test revealed a significant difference between On-Button and EarWiggl'In, with the difference of 36.28[ms] among the average input time. No significant differences were observed in the other combinations. The results suggest that the input speed of ear wiggling is comparable to, but slightly slower than finger-pressed input.

The NASA-RTLX scores from Task A are shown in Fig. 7. No significant differences were found between methods using the Wilcoxon test. However, when examining the average scores there was not much difference in Temporal demand (TD), likely due to the inherent workload of Task A. Physical demand (PD) and effort (EF) scores for EarWiggl'In scored higher compared to the other methods. Additionally, the other subscale scores for EarWiggl'In were slightly higher than the other methods. This can be attributed to the novelty of using auricular muscles, as they are vestigial and ear wiggling is not typically used in daily life. With regular use, the total workload may decrease over time. Overall, the total workload score for EarWiggl'In was slightly higher than those for the other methods.

Fig. 8 shows the recorded waveforms for the left and right sensors during each gesture in Task B. The mean waveform for each

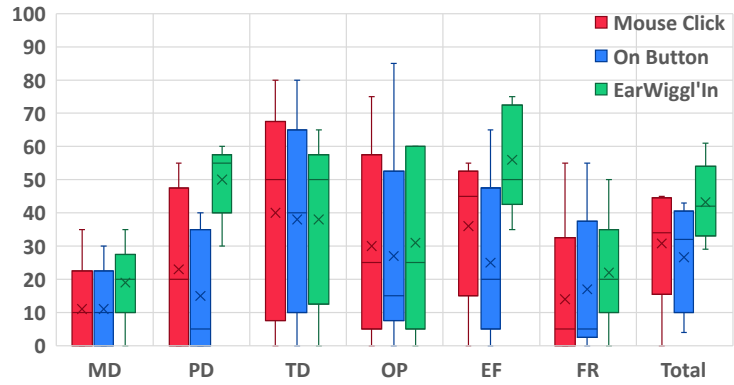


Figure 7: NASA-RTLX scores between Mouse Click, On-Button, and EarWiggl'In. (MD: Mental Demand, PD: Physical Demand, TD: Temporal Demand, OP: Performance, EF: Effort, FR: Frustration)

gesture is highlighted in red. Note that no mean waveform is shown for “mash wiggle”, due to variations in pressing intervals between participants. The data were organized by adjusting the rising point at 0.16[s], applying a windowing length of 1.28 [s], and subtracting the minimal value. The gestures “2 / weak 2 wiggle” can see two peaks and a trough in between. The mean waves of these gestures are slightly attenuated compared to the actual waveforms due to variations in intervals among participants. The gestures “1 wiggle / 2 wiggle / weak 1 wiggle / weak 2 wiggle” demonstrated similar amplitude values across the two sensors. In contrast, for “Left / right wiggle”, the opposite sensor (e.g. the left sensor during a “right wiggle”) shows higher amplitude than the other. This occurs because the glasses shift slightly to the side being wiggle, causing the opposite sensor to press more firmly against the nose. The average holding time for “hold wiggle” was 2005.6[ms](±157.7[ms]), which closely matched the instructed duration. For the “mash wiggle”, participants achieved an average wiggle rate of 3.12 times per second.

A significant difference was observed between the amplitude of “1 wiggle” and “weak 1 wiggle”. This is more evident in Fig. 9, which shows the amplitude distribution for gestures among participants. The line in each graph represents a least squares regression calculated from “1 wiggle” and “weak 1 wiggle”. The slope indicates which ear was dominant for each participant. P1 had a greater dominance with his right ear, P2 with his left ear, while other participants demonstrated relatively equal strength between their ears. All participants show a clear distinction in the distribution between “1 wiggle” and “weak 1 wiggle”. The distribution of “2 wiggle” overlapped with “1 wiggle” among P2, P3, P4, and P5. The distribution of “weak 2 wiggle” overlapped with “weak 1 wiggle” among P2, P4, and P5. The distribution of “left wiggle” and “right wiggle” was separated among P1, P2, P3, and P5. This result aligns with what P2 and P5 mentioned about capability of wiggling their ears independently and P1 mentioning he was capable in the past. Surprisingly, the results from P3 were also distinguishable, despite he did not report this ability.

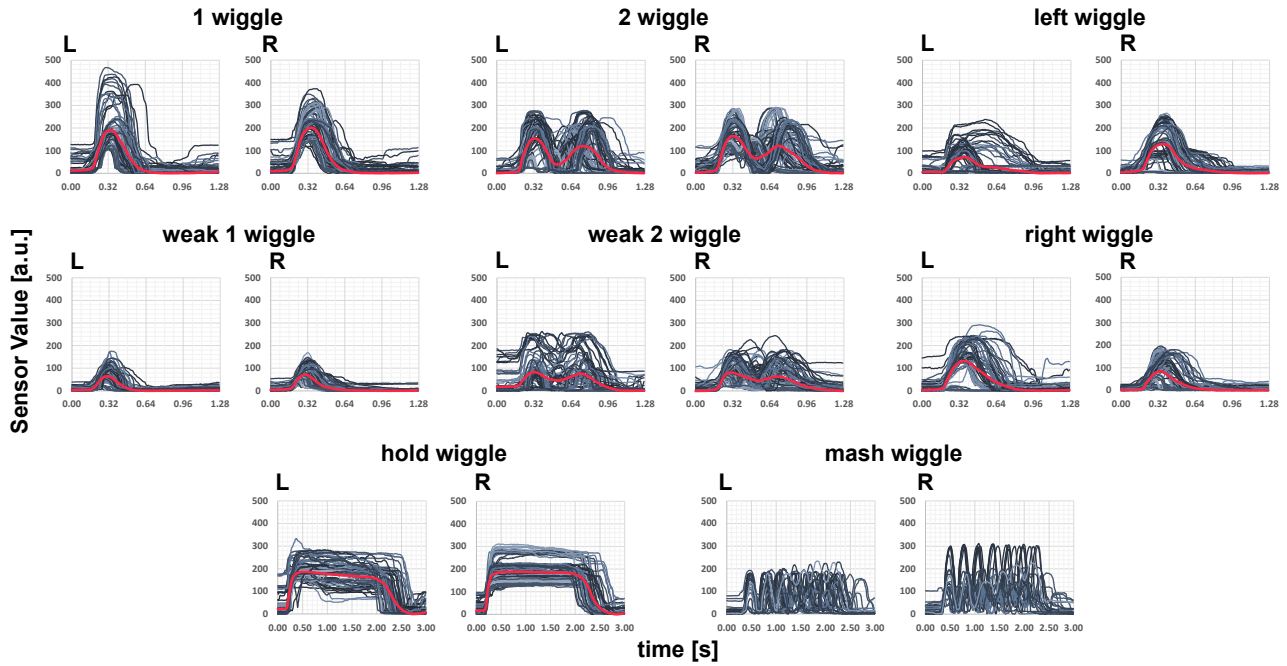


Figure 8: The recorded waveforms and the mean waveforms from left and right sensors of each gesture. Each mean waveform is colored red and was generated from 75 samples (15 samples per participant). The waveforms were organized by adjusting the rising point at 0.16[s], applying a windowing length of 1.28[s], and subtracting the minimal value.

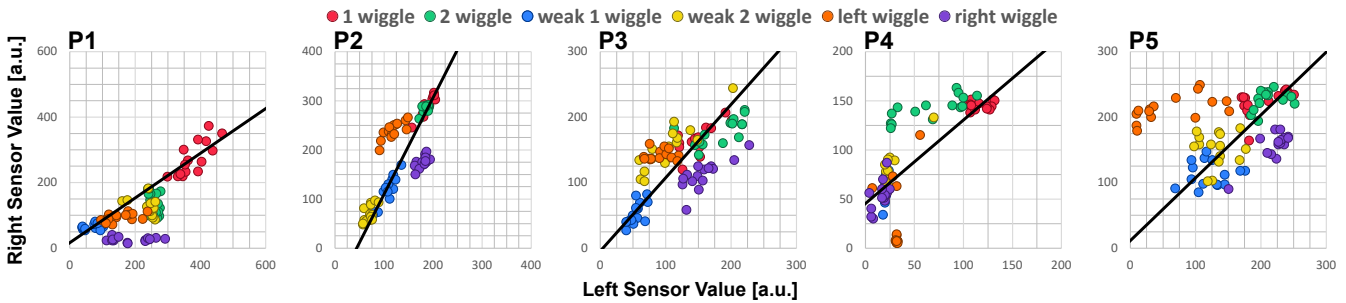


Figure 9: The distribution of amplitude among gestures. The line shown in the graph is drawn by the least squares method using the values of “1 wiggle” and “weak 1 wiggle”.

However, the distribution of “2 wiggle” and “weak 2 wiggle” overlapped for P1 and P3, while the distribution of “left / weak 1 wiggle” overlapped for P1. Additionally, the distribution of “left / right / weak 1 wiggle” overlapped for P4. The distribution of P4 was non-linear compared to those of the other participants. This non-linearity is likely due to variation in the contact between the user and the sensor, which reflects the non-linear resistance decrease of the pressure sensors.

The scores of the NASA-RTLX from Task B are shown in Fig. 10. No significant differences were found between any comparison using the Wilcoxon test. “1 wiggle” scored the lowest across multiple subscales, followed closely by “Weak 1 wiggle”. “2 wiggle” scored slightly higher than “1 wiggle” and “weak 2 wiggle” scored slightly higher than “weak 1 wiggle”. In general, the weak gestures scored higher compared to the normal gestures in several subscales. “Left

wiggle” scored higher than “right wiggle” on all subscales, except temporal demand (TD). This result shows a bias between the two gestures. “Hold wiggle” and “mash wiggling” recorded high scores in the physical demand (PD) and effort (EF). The frustration (FR) subscale also scored very high for “mash wiggling”. These results suggest that “left / right / hold wiggle” and “mash wiggling” were more challenging due to their complexity.

In the questionnaire after the tasks, participants reported experiencing a slight tiredness around the ear, but no pain. Regarding the inhibition of other movements, using the proposed method P1 felt mildly inhibited, while P4 and P5 reported they didn’t feel much inhibited and the remaining participants did not perceive any inhibition. When asked about the practical utility of EarWiggl’In, P1, P2, and P4 answered that it would be highly useful for daily device operation, while P3 and P5 considered it useful. Additional

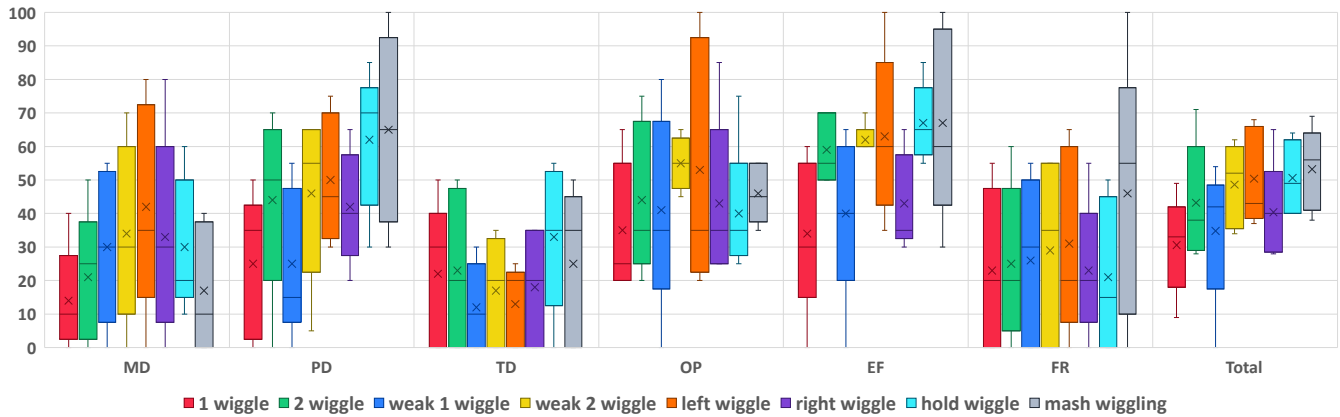


Figure 10: NASA-RTLX scores of each gesture(MD: Mental Demand, PD: Physical Demand, TD: Temporal Demand, OP: Performance, EF: Effort, FR: Frustration)

comments included P1 finding “mash wiggling” particularly challenging. P4 described the experiment as enjoyable and hoped it would be practically applied. P5 stated the gestures felt more like the nose pads pressing against the nose rather than pulling the glasses back using the ears.

6 Discussions & Future Work

Task A results showed a significant difference between On-Button and EarWiggl’In, with an average input time difference of 36.28[ms]. This difference is relatively small in practical scenarios because, unlike conditions of Task A, the On-Button would require additional time for users to reach for and aim at the button located on the frame of the glasses. This added action could increase the input time and potentially result in misoperations. In contrast, EarWiggl’In maintains constant contact with the user while the device is worn, eliminating the need for reaching or aiming. Although ear wiggling is less commonly used compared to finger movements, a long-term use of EarWiggl’In could enhance input speed, potentially making it faster than reaching a button.

Task B demonstrated ear wiggling is achieved by analyzing the rise and fall of the sensor values, and gestures can be classified by their amplitude, number of peaks, and duration. The different distribution patterns were seen among participants, likely reflecting user characteristics. As a limitation, the number of participants was limited, due to the fact that currently 10–20% people are estimated being able to ear wiggling [8, 14, 17]. Further research should construct a training method to help individuals regain control of their movement, thereby expanding number of potential users. Biofeedback has been shown to be an effective training method in previous studies [6, 16]. Further investigation is needed to determine whether these training methods can be applied to EarWiggl’In.

Other human movements have been observed to affect the sensor value, such as: wearing the device, blinking, walking, heartbeat, and changes in head direction. Classification and evaluation of these effects has been incomplete. However, distinct value changes have been observed for certain gestures. For instance, blinking causes the sensor value to decrease from the base value. These effects

may be removable as the effect of the wearing position has been removed by subtracting the minimal value in Fig. 8. The effects of walking and changing head direction may be removable using accelerometers and gyroscopes. Future research should focus on classifying these movements, enabling their use as additional input methods or for human monitoring in smart glasses.

As a new user interface, a long-term study is necessary to fully evaluate the practicality and usability of EarWiggl’In. During the user study, the participants reported that they had never experienced pain during or after ear wiggling, although they did feel a slight tiredness after the user study. Further research is required to determine whether this tiredness diminishes with continued use or if it represents a significant limitation.

For applications, EarWiggl’In can be used as an input method for target selection. When integrated into smart glasses equipped with gaze input detection, it enables hands-free operation using a single device. This system serves as an effective interface for people with spinal injuries, as the muscles controlling ear wiggling do not pass through the spinal cord. Taking photos with camera-equipped smart glasses may be practical, allowing you to take a photo instantly, faster than raising your hand to the controller. Shortcuts can be applied such as controlling a music player, taking screenshots, or switching virtual desktops. It could also be used for entertainment purposes, such as animating cat ears on virtual avatars through ear wiggling.

7 Conclusion

EarWiggl’In demonstrated that ear wiggling can be effectively detected by pressure sensors embedded on the nose pads, providing a hands-free input. This system offers fast and voluntary input comparable to a mouse click, without interfering with daily activities. The pressure sensors ensure environmental resistance and a lightweight design, making it practical for wearable applications.

In the user study, the input speed of EarWiggl’In was found to be equal to or slightly slower (by 36.28[ms]) than pressing a button with a finger. However, this small difference is expected to diminish in practical scenarios and with long-term use. While NASA-RTLX

scores indicated higher physical demand and effort for EarWiggl'In compared to other methods, while other subscale scores were only slightly elevated. The study also revealed significant variations in gestures, including differences in amplitude, number of peaks, and duration depending on the gesture. For example, “1 wiggle” and “weak 1 wiggle” showed distinct amplitude differences, while “left wiggle” and “right wiggle” exhibited higher sensor values on the opposite side due to the movement. NASA-RTLX scores for the gestures highlighted differences in difficulty, with “1 wiggle” and “weak 1 wiggle” being the easiest and “mash wiggle” being the most challenging.

These findings suggest that EarWiggl'In can be effectively utilized for applications such as gaze cursor target selection or photo capture gestures for smart glasses. With future development, EarWiggl'In could become a practical daily input method for smart glasses.

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